EXATHON INDIA 2026 · TRACK 01: MONEY, RISK & REGULATION

Reconciliation and Fraud Detection at Warehouse Speed

One explainable, set-based SQL pipeline in Exasol that closes the books and flags fraud across 1,000,000 transactions.

1M
transactions reconciled

98.8%
fraud precision (HIGH)

~2 s
to load 1M ledger rows

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THE PROBLEM

Money leaks in the gap between systems

Banks and fintechs still match ledgers to bank statements by hand, while fraud is caught after settlement.

Days

Manual reconciliation

Ledger and bank statement are matched in spreadsheets; cycles stay open for days.

Hidden

Mismatches & duplicates

Missing entries, amount variances and double postings quietly sit on the books.

Late

Fraud detection

Velocity bursts and account takeovers are found after the loss has occurred.

Research question: can both tasks run as a single, explainable SQL pipeline inside an analytical database at 1M-transaction scale?

SYSTEM DESIGN

Everything happens inside Exasol

No dataframes, no ETL framework. Data in, SQL out.



METHODOLOGY

Three-stage matching, explainable scoring

Pure SQL: joins, window functions and aggregates over 1M rows.

Reconciliation

1. **Reference match** on transaction ref; classify exact vs. amount mismatch; rank duplicate postings with `ROW_NUMBER`.
2. **Fuzzy match** when the bank ref is missing: account + amount + value date within 2 days, one-to-one.
3. **Exceptions:** missing in bank, unmatched bank, duplicate posting.

Fraud risk score (0–100)

Additive signals, every alert ships with reasons.

- **Velocity burst ≤ 45**
- **Amount vs. norm ≤ 50**
- **Foreign city 35**
- **Rare device 15**
- **Foreign + rare device +15**
- **Night high value 10**

HIGH ≥ 60 **MEDIUM** ≥ 35 **LOW** below

Why an analytical database fits this job

Heavy self-joins and window functions are the workload Exasol is built for.

~2 s
to bulk-load 1,000,000 ledger rows

0
dataframes or ETL frameworks in the matching path

10-min
velocity window computed by a set-based self-join

4
commands to reproduce the full pipeline

Runs locally on Exasol Personal; the same SQL scales to larger deployments unchanged.

RESULTS I · RECONCILIATION

95% of the ledger closes automatically

92.5% exact by reference, plus 2.5% recovered by fuzzy matching.



Matched 924,812 Fuzzy 24,918 Amount mismatch 30,135 Missing in bank 20,135

VALUE SURFACED (BASE CURRENCY, INR)

57.3M
missing in bank
20,135 txns

27.5M
unmatched bank credits
15,240 rows

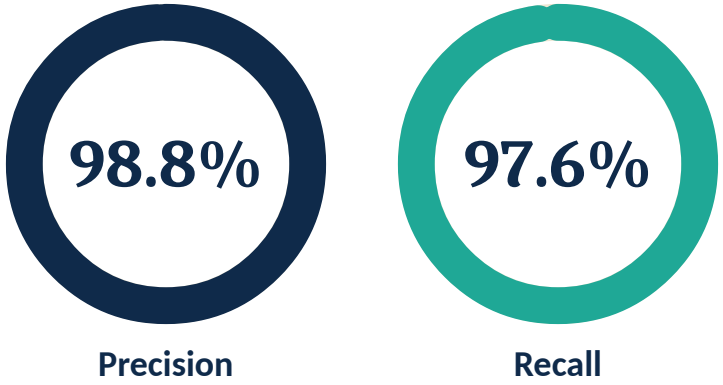
26.5M
duplicate postings
9,559 rows

0.77M
amount variances
30,135 txns

RESULTS II · FRAUD DETECTION

98.8% precision at 0.79% alert rate

HIGH band: 7,898 of 1,000,000 transactions flagged, so analyst workload stays small.



Recall by fraud pattern

Account takeover	2,800 / 2,800 · 100%
Amount spike	1,966 / 2,000 · 98.3%
Velocity burst	3,038 / 3,200 · 94.9%

TP 7,804 · FP 94 · FN 196 | Example alert: 49,977 in Kyiv, night-time, rarely used device, far above customer norm → score 100.

RIGOR

Limitations and next steps

Stated upfront: these are controlled-benchmark results.

Threats to validity

- **Synthetic data and labels:** fraud patterns were injected by us.
- **Optimistic thresholds:** designed knowing the injected patterns; no held-out split.
- **Narrow coverage:** three typologies; adversarial or slow-drift fraud not modeled.
- **Rule-based:** auditable, but not learned from data.

Future work

- Validate on anonymized real payment data with independent labels.
- Tune thresholds on train/validation splits; compare learned models run in-database.
- Audit rules for geographic and demographic bias (e.g., the foreign-location signal).
- Streaming ingestion for true real-time alerting.

REPRODUCIBLE IN FOUR COMMANDS

Try it yourself

```
$ exasol install local
$ python src\load_fast.py
$ python src\run_sql.py sql\02_reconcile.sql sql\03_fraud.sql
$ python src\show_results.py
```

github.com/sisodiyaanand/exathon-recon · MIT license

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